

RESEARCH ARTICLE

Cozie Singapore: Scalable crowdsourced smartwatch micro-surveys to capture urban heat and noise perception and behavior

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ARTICLE HISTORY

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ABSTRACT

[PLACEHOLDER: 150–250 word structured abstract. Context (urban sustainability and the human experience of everyday space types); the ORENTH / Cozie-Singapore watch-survey dataset (9,962 responses, 106 participants); the aim (characterize how perception, physiology and ambient weather differ across space types — Home, Office, Classroom, Other Indoor, Outdoor, Transportation); the approach (exploratory analysis + effect-size-based statistical characterization + weather augmentation + discriminative modeling); the headline finding (function/behaviour separates spaces, but ambient weather barely marks space while genuinely tracking thermal preference); and the implication for sustainable urban design.]

KEYWORDS

[PLACEHOLDER: 4–6 keywords — e.g. urban sustainability; thermal comfort; wearable sensing; space-type characterization; watch-survey; Singapore]

1. Introduction

[PLACEHOLDER: Opening framing — why the everyday human experience of urban space types matters for urban sustainability (health, comfort, energy, liveable cities).]

Motivate the problem. Cities in the tropics, indoor/outdoor thermal experience, the role of air-conditioning, and why understanding how people perceive and physiologically respond to different space types is relevant to sustainable urban development.]

1.1. Crowdsourcing humans-as-a-sensor data across the urban context

Review prior work on (i) urban thermal comfort and microclimate, (ii) wearable / longitudinal in-situ sensing and micro-surveys (Cozie), (iii) space-type or context characterization, and (iv) streetscape/urban-form exposure. Identify the gap this study fills.

[PLACEHOLDER: State the aim and research questions, e.g.: RQ1 — How do perception, physiology and ambient weather differ across space types? RQ2 — Does ambient weather help identify a space type? RQ3 — Does wanting “cooler” track actual heat? List the contributions and give a short roadmap of the paper.]

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2. Methods

[PLACEHOLDER: One-paragraph overview of the methodological pipeline from raw watch-survey data to characterization and modeling.]

2.1. Crowdsourced watch-based micro-survey and physiological and environmental data

[PLACEHOLDER: Describe the ORENTH / Cozie-Singapore field study — deployment, smartwatch micro-surveys, participants, study period, and Singapore urban context. Note participant recruitment and ethics as appropriate.]

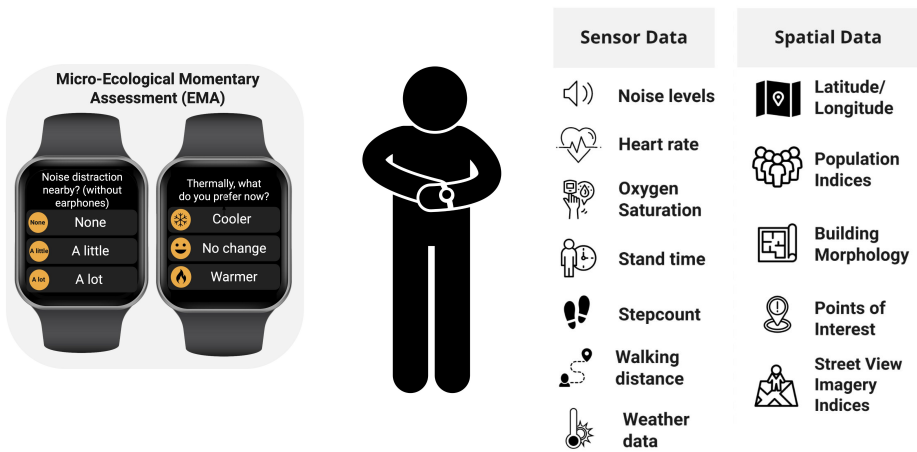


Figure 1. [PLACEHOLDER: Study design and data-collection workflow.]

[PLACEHOLDER: Describe the derived survey-level dataset (`cozie_sg_surveys_with_health_aggregates`): 9,962 responses, 106 participants, 99+ columns. Explain how HealthKit physiological metrics are aggregated to a 1-hour pre-survey window and how home locations were recovered.]

2.1.1. Micro-survey structure

[PLACEHOLDER: List subjective variables — thermal preference, noise level, noise kind, earphones, social context (alone/group), activity — and their response categories.]

[PLACEHOLDER: Describe 1-hour aggregates — mean heart rate, total steps, walking distance.]

2.1.2. Urban contextual indicators using Urbanity

[PLACEHOLDER: Describe geolocated environmental features — green/sky/building/ road view, visual complexity, building count, population, POI categories.]

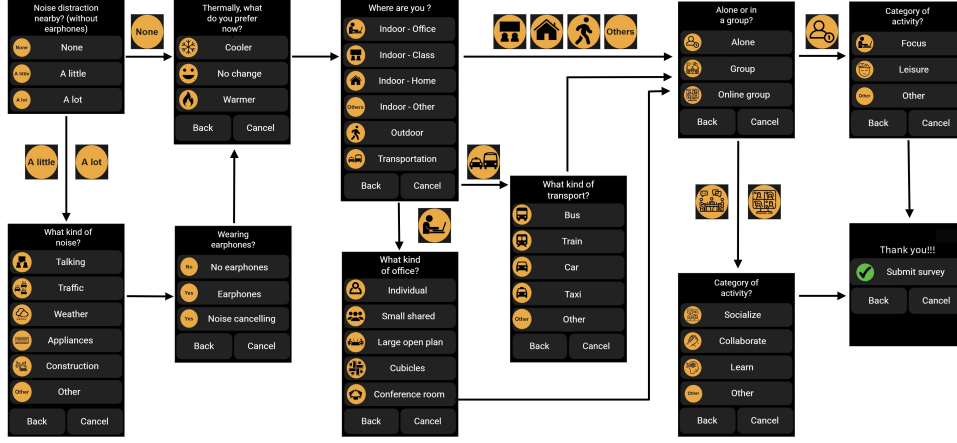


Figure 2. [PLACEHOLDER: Cozie smartwatch micro-survey interface and response flow.]

2.1.3. Weather data

[PLACEHOLDER: Describe the nearest-station weather aggregation — air temperature, relative humidity, heat index, wind speed, rain, is-raining flag — units and coverage (92–96%). Note this is ambient, not skin-level, exposure.]

2.2. Data processing and characterization

[PLACEHOLDER: Describe the EDA strategy from notebook 01 — univariate survey / health distributions, thermal relationships, temporal profiling in Singapore local time, and geospatial mapping.]

[PLACEHOLDER: Describe the characterization methods from notebook 03 — perception fingerprints (row-normalized conditional proportions and lift), association and effect size (Kruskal–Wallis ϵ^2 , chi-square / Cramér’s V, Spearman ρ), and multivariate structure (MCA + k-modes clustering).]

[PLACEHOLDER: Describe how weather features augment the characterization and the direct test of whether thermal preference tracks actual heat index / rain.]

[PLACEHOLDER: Describe the space-type classifier (features = perceptions $\pm$ weather), validation, and importance / SHAP attribution.]

3. Results

3.1. Dataset overview and survey responses

[PLACEHOLDER: Summarize the dataset and headline distributions of survey responses and health metrics.]

[PLACEHOLDER: Describe the survey response-flow structure.]

[PLACEHOLDER: Describe the island-wide geospatial patterns — response density, spatial thermal preference, green-view exposure, heart-rate hotspots, and the per-space-type density facets.]

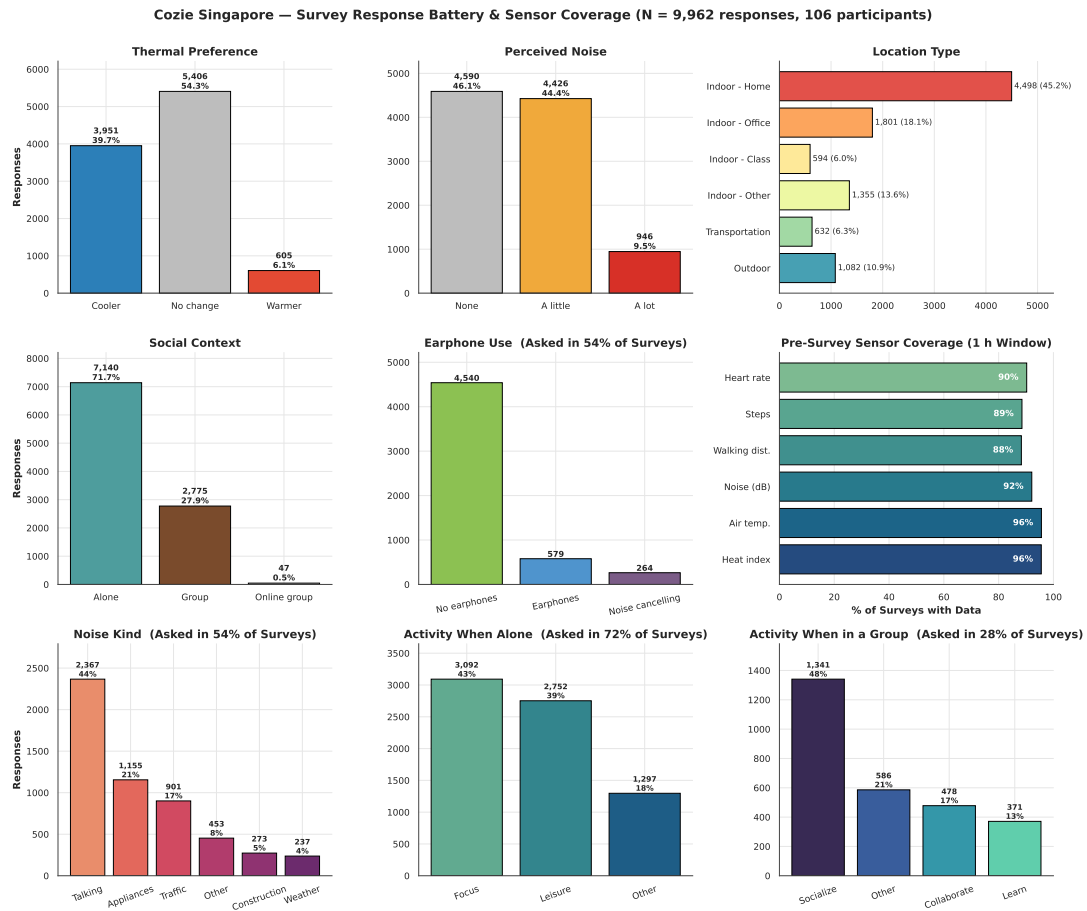


Figure 3. [PLACEHOLDER: Survey response battery, conditional follow-ups, and linked-sensor coverage (Figure A1).]

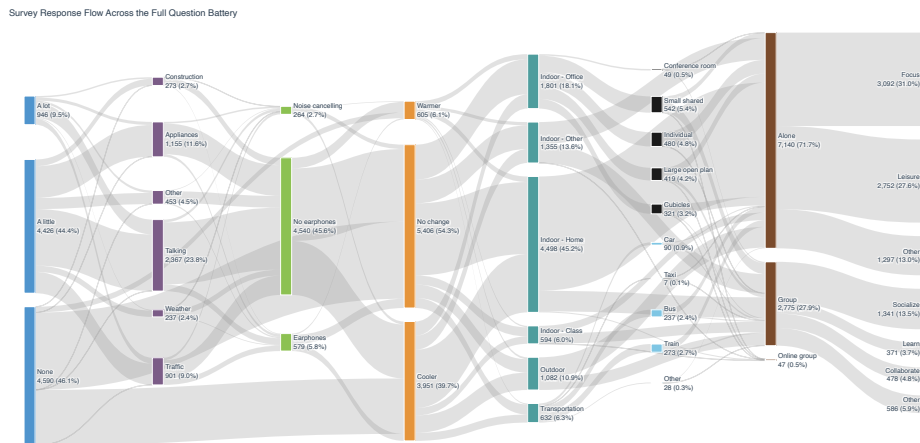


Figure 4. [PLACEHOLDER: Conditional response flow through the watch-survey questionnaire.]

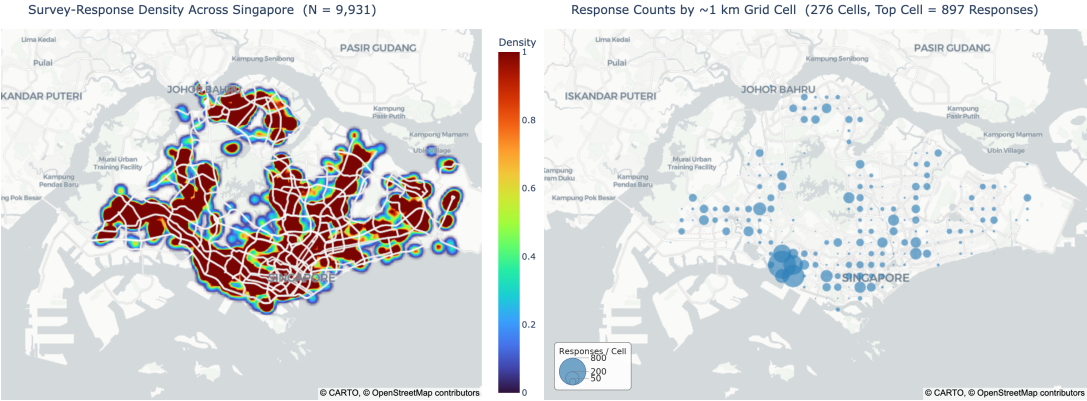


Figure 5. [PLACEHOLDER: Survey-response density and response counts by approximately 1-km grid cell across Singapore (Figure B4).]

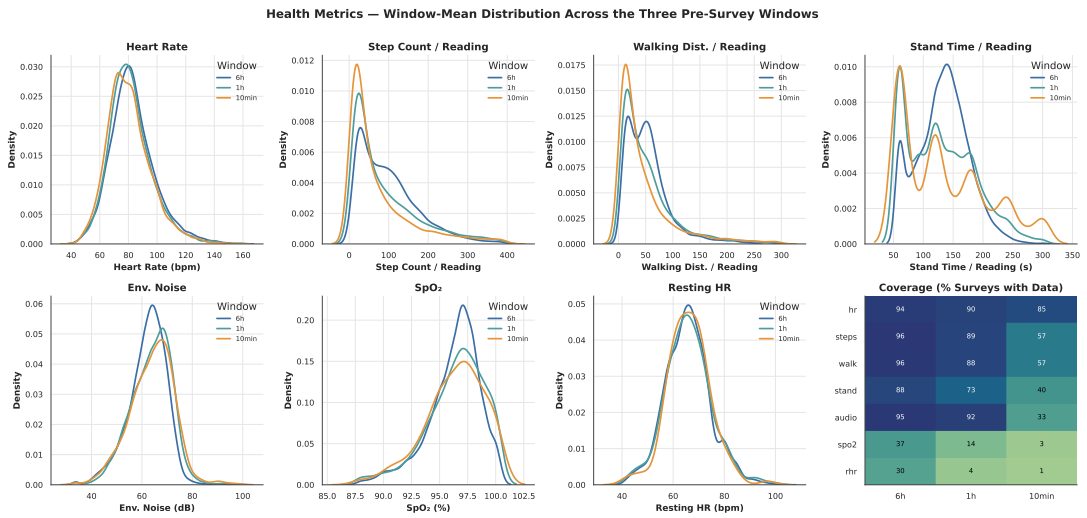


Figure 6. [PLACEHOLDER: Wearable-health metrics across pre-survey aggregation windows (Figure A2).]

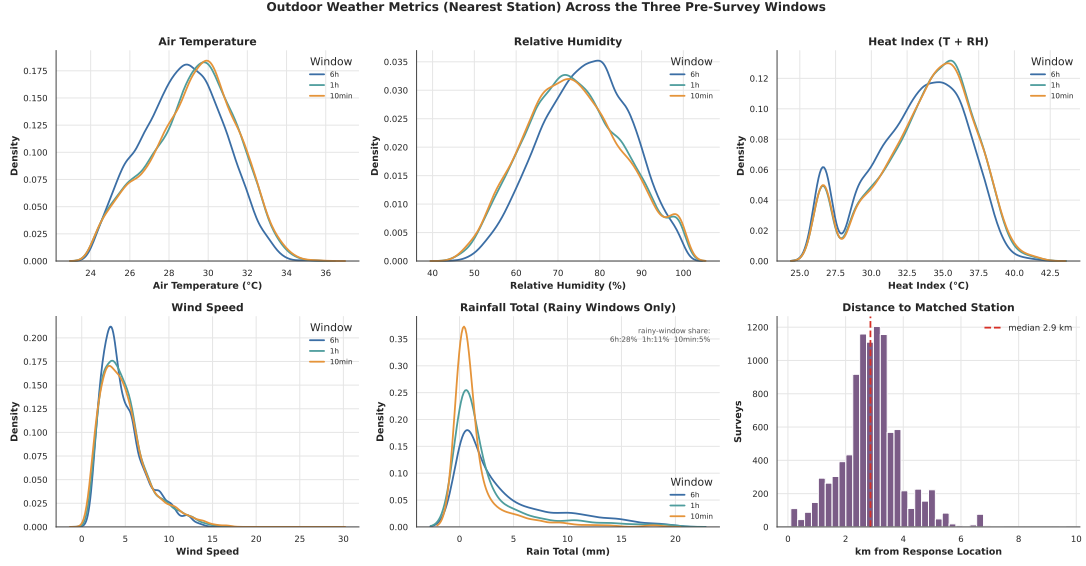


Figure 7. [PLACEHOLDER: Nearest-station outdoor ambient weather across pre-survey aggregation windows (Figure A3).]

3.2. Temporal patterns of response submission

[PLACEHOLDER: Describe when surveys are filled out (Singapore local time) and how timing is strongly space-type dependent (Office/Classroom weekday-afternoon vs Home evening/weekend vs bimodal Transportation).]

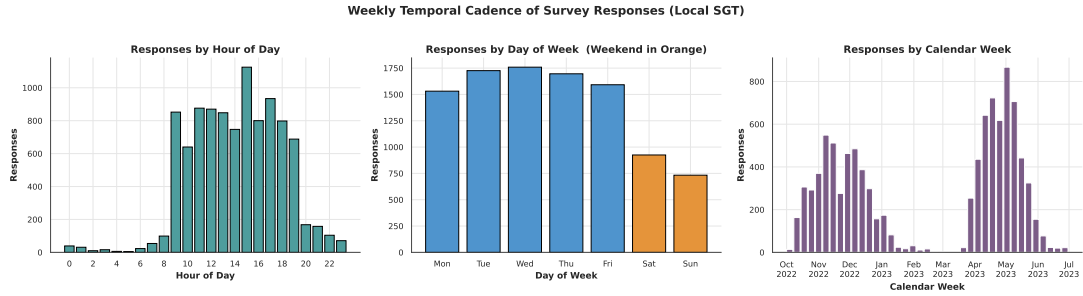


Figure 8. [PLACEHOLDER: Weekly temporal cadence of survey responses in Singapore local time (Figure D2).]

3.3. Per-participant heterogeneity

[PLACEHOLDER: Describe between-participant variation in space-type / perception shares (stacked bars and distribution-of-shares histograms).]

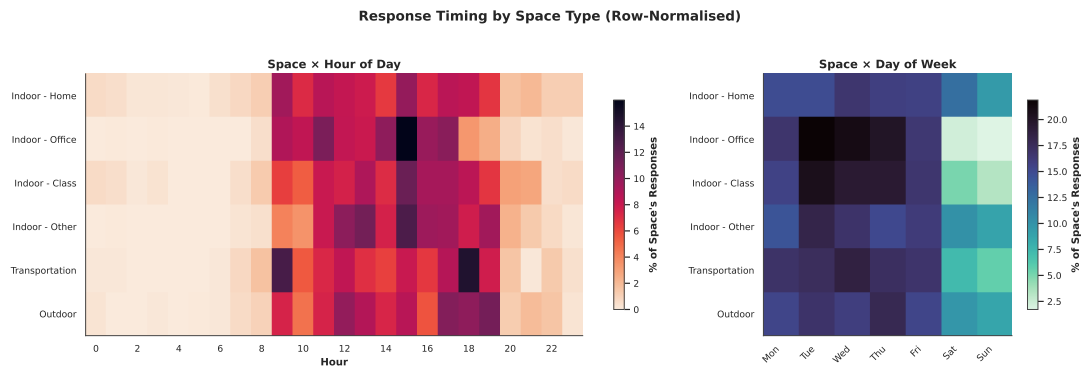


Figure 9. [PLACEHOLDER: Response timing by space type, shown by hour of day and day of week (Figure D4).]

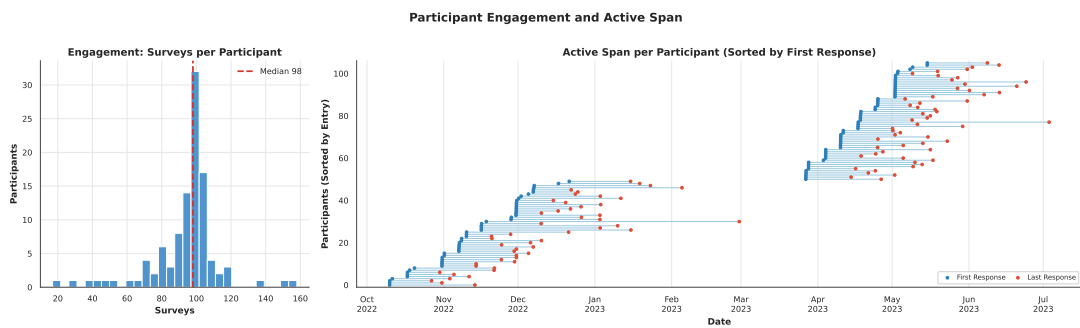


Figure 10. [PLACEHOLDER: Participant engagement and active survey span (Figure E2).]

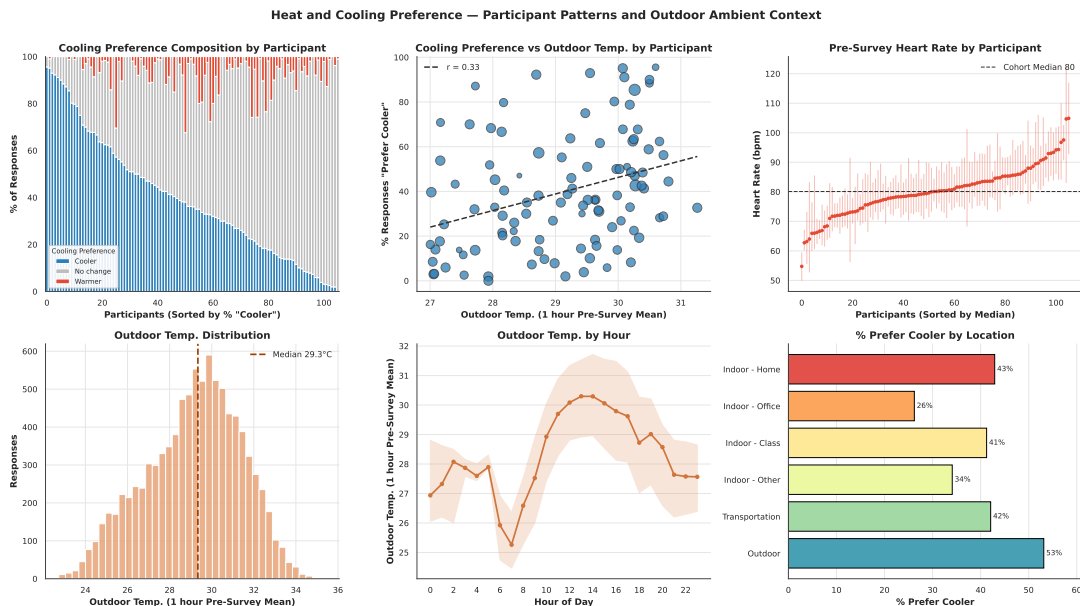


Figure 11. [PLACEHOLDER: Participant-level and contextual patterns in cooling preference (Figure F2).]

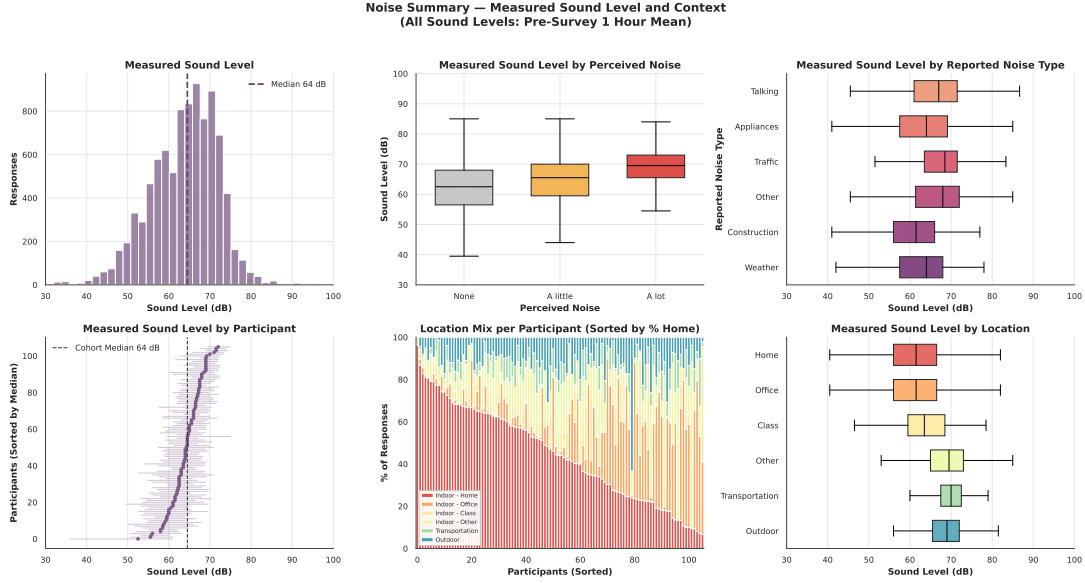


Figure 12. [PLACEHOLDER: Measured sound level across subjective, participant, and location contexts (Figure G1).]

3.4. Cooling preference

3.5. Noise preference

3.6. Perceptual fingerprints by space type

[PLACEHOLDER: Describe the perception fingerprint ($P(\text{response}|\text{space})$) and the lift table flagging distinctive perceptions per space type.]

3.7. Statistical association and effect size

[PLACEHOLDER: Report association strength between space type and each perceptual variable (Cramér's V), emphasizing effect sizes over p-values.]

3.8. Multivariate structure of experience

[PLACEHOLDER: Describe MCA + k-modes clustering of the subjective experience and how clusters relate (or not) to space type.]

3.9. Physiological and weather signatures

[PLACEHOLDER: Report physiological gradient across spaces (sedentary Home to mobile Outdoor/Transport; steps largest effect) and the weak ambient-weather signatures — ambient weather barely marks space type; Outdoor is not the hottest.]

3.10. Weather versus thermal preference

[PLACEHOLDER: Report the key new result — “Cooler” responses occur in genuinely hotter (higher heat-index) and drier conditions; rain suppresses cooling demand. Effect

is real but modest.]

3.11. Outdoor environmental characterization

[PLACEHOLDER: Restricted to Outdoor responses — streetscape/built-form profile, its (weak) link to thermal preference, environment–physiology nulls, and actual weather vs outdoor thermal preference.]

3.12. Discriminative modeling of space type

[PLACEHOLDER: Report the space-type classifier performance and feature importance (activity and noise-kind as strong space descriptors; ambient weather adds little).]

4. Discussion

4.1. Principal findings

[PLACEHOLDER: Synthesize: function/behaviour (activity, noise + coping, movement) separates space types robustly; thermal, environmental and physiological expectations are mostly weak or convergent nulls; the weather correction shows ambient weather is nearly useless for identifying a space yet genuinely drives outdoor thermal preference.]

4.2. Interpretation against intuition

[PLACEHOLDER: Discuss the reinforced vs contradicted intuitions (e.g. Outdoor is not the hottest; streetscape greenery does not move thermal preference; actual heat does).]

4.3. Implications for sustainable urban design

[PLACEHOLDER: Connect findings to the journal scope — what they mean for thermal-comfort-aware, health-supportive and energy-conscious urban design and policy in tropical cities.]

4.4. Limitations

[PLACEHOLDER: Participant nesting / pseudoreplication; ambient nearest-station weather as a coarse proxy for personal exposure; convenience sample; Singapore-specific A/C context; self-reported perceptions.]

4.5. Future work

[PLACEHOLDER: Directions — personal microclimate sensing, larger/more diverse cohorts, cross-city replication, mixed-effects modeling accounting for participant nesting.]

5. Conclusion

[PLACEHOLDER: Concise restatement of the contribution and headline takeaway for sustainable urban development.]

Acknowledgements

[PLACEHOLDER: Acknowledgements.]

Disclosure statement

[PLACEHOLDER: Declare any competing interests.]

Funding

[PLACEHOLDER: Funding sources.]

Data availability statement

[PLACEHOLDER: Statement on ORENTH / Cozie-Singapore data availability.]

References